

Museum in the Classroom: Engaging Students with Augmented Reality Museum Artifacts and Generative AI

Carina Ly
Computer Science Department
Stanford University
Stanford, California, USA
carinaly@stanford.edu

Eleanor Peng
Computer Science Department
Stanford University
Stanford, California, USA
eleanor.peng@stanford.edu

Katie Liu
Computer Science Department
Stanford University
Stanford, California, USA
katiel25@stanford.edu

Anthony Qin
Computer Science Department
Stanford University
Stanford, California, USA
antqin@stanford.edu

Grace Howe
Archaeology Center
Stanford University
Stanford, California, USA
ghowe25@stanford.edu

Alan Y. Cheng
Computer Science Department
Stanford University
Stanford, California, USA
ayc@stanford.edu

Andrea Cuadra
Computer Science
Olin College
Needham, Massachusetts, USA
acuadra@olin.edu

Abstract

Museum field trips provide a rich learning experience for children. However, they are complex and expensive for teachers to organize. Fortunately, digitization of museum artifacts makes it possible to use museum resources within the classroom. Museum in the Classroom (MITC) explores how augmented reality (AR) and generative artificial intelligence (AI) can create an interactive learning experience around museum artifacts. This iPad app allows educators to select historical topics from a curated artifact library, generating AR-based exhibits that students can explore. MITC engages students through interactive AR artifacts, AI-driven chatbots, and AI-generated quiz questions, based on a real exhibition at the Cantor Arts Center at Stanford University. A formative study with middle schoolers ($N = 20$) demonstrated that the app increased engagement compared to traditional learning methods. MITC also fostered a playful and comfortable environment to interact with museum artifacts. Our findings suggest that combining AR and AI has the potential to enrich classroom learning and offer a scalable alternative to traditional museum visits.

CCS Concepts

• **Human-centered computing** → *Ubiquitous and mobile devices*; Text input; **Activity centered design**; *Interactive systems and tools*; **Mobile devices**; **Mixed / augmented reality**; • **Applied computing** → *Interactive learning environments*.

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Keywords

Education/Learning; Children/Parents; Artifact or System

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1 Introduction

Museums are important institutions for children's learning [4], and field trips to museums are a common activity in children's classrooms. However, field trips are complex to organize, expensive to run, and require significant work from the teacher [17, 27]. To tap into the wealth of resources that museums provide, we could also bring parts of these museums to the classroom, which has been found to increase student engagement [33]. As more museums digitize their collections, classrooms now have greater access to authentic objects and materials curated by experts. These digitized museum objects serve as authentic source materials that teachers can integrate into their lessons. But could you create a "museum in the classroom" that brings museums to life the way that museums do?

While the museum experience itself is irreplaceable, there are several ways to leverage emerging technologies towards realizing this vision. In particular, we can utilize augmented reality (AR) technologies to bring digital museum artifacts from the screen into the real world. We could also investigate how generative AI, large language models (LLMs) in particular, could be used in tandem with these AR artifacts support students to engage with those artifacts in a more personalized and interactive manner.

To explore these avenues, we have built Museum in the Classroom (MITC), an tablet-based app designed for the classroom that creates AR-based museum exhibits. Students can visualize 3D digital artifacts in AR, providing a more immersive learning experience than reading from a textbook. A built-in note panel in the AR experience allows students to document observations and receive AI-driven feedback to support their learning process. The app also includes supplementary features that is not generally offered in traditional learning methods, such as generated artifact descriptions that suitable for students' reading levels, trivia-like reinforcement questions, and a chatbot that offers immediate answers to student inquiries. In this study, we investigate two main research questions:

- RQ1:** What are opportunities and challenges of combining augmented reality and generative AI to bring elements of a museum experience into the classroom?
- RQ2:** How do middle school students interact with this technology in the history classroom? What do they like and dislike, and how do they compare it to other experiences?

2 Related Work

2.1 Museum Learning in the Classroom

A wide body of research has investigated the benefits of museum field trips on both cognitive (e.g., [3]) and affective (e.g., [26]) learning outcomes for children. Museum artifacts, and objects in general, come with rich contexts that support student sensemaking through direct experience [1]. However, a teacher must spend a high level of time and effort to maximize learning outcomes [17, 27], including preparation before the trip, orienting the learning during, and facilitating reflection after [5]. One alternative to museum field trips is to instead make use of digitized museum artifacts, which not only engage students [7], but they also give teachers the flexibility to decide how and when to integrate museum resources into their lesson plans [14]. Our work therefore focuses on museum artifacts as a vehicle for learning, and we partnered with a local art museum to digitize a set of historical artifacts that we used in our prototype.

2.2 Mobile Augmented Reality for History Education

Augmented reality (AR) technologies superimpose virtual objects onto a person's visual field [8]. By positioning learners in a real-world context, AR can engage learners' spatial cognition [24] and facilitate situated learning [19, 31]. In the fields of history and heritage education, AR can bridge the past and the present by immersing students in historical events and contexts [6]. There has been much work on using AR to augment the learning experience in situ at historical sites (e.g., [9, 25]), as well as at museums [16, 34], to create more engaging and immersive learning experiences. However, there has been comparatively less research done on how AR can be used to support in-classroom learning. Therefore, our approach with MITC is to use AR to bring digital museum artifacts to life within a classroom context.

2.3 Large Language Models for Student Learning

Large language models (LLMs), with their capability to understand and generate human text, have great potential to support students in learning. For example, recent work has found LLMs to be effective at generating understandable explanations [20], and having students interact with an LLM as part of a learning experience can help keep them engaged and on task [10]. Additionally, LLMs are capable of generating high-quality useful questions for students to assess their understanding of concepts [13]. Chatbots are an especially popular use case for LLMs, and they have been tested across a wide range of education applications (e.g., [11, 21, 29]), and they are typically simple to create with services that provide access to LLMs. In this work, we make use of an LLM in a variety of ways to support the overall goal of getting students to engage with museum artifacts, from quiz question generation to a chatbot that students can discuss with and ask questions to.

2.4 Artifacts Related to Physical Proxies

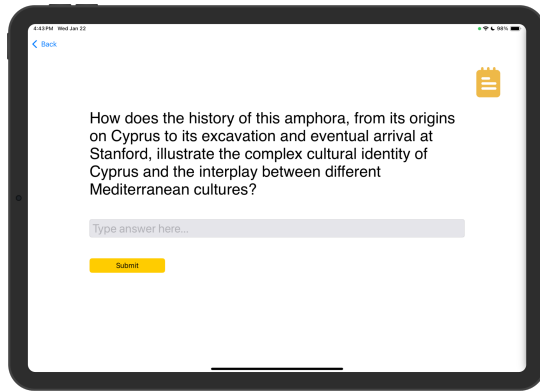
Recent research has explored the concept of opportunistic interfaces, where semantically matching virtual content is associated with physical objects. This type of technology allows users to transform everyday objects into interactive surfaces, expanding the possibilities for engaging learning experiences through tangible interactions [12, 32]. In educational settings, the ability to directly interact with physical proxies enhances engagement and understanding. One use case could be for artifacts, in which they can be projected onto everyday classroom objects, allowing students to explore historical objects in greater depth without the need for field trips. Research has demonstrated that tangible interaction improves cognitive engagement and retention, making it an effective tool for subjects that benefit from spatial understanding [18].

3 Museum in the Classroom

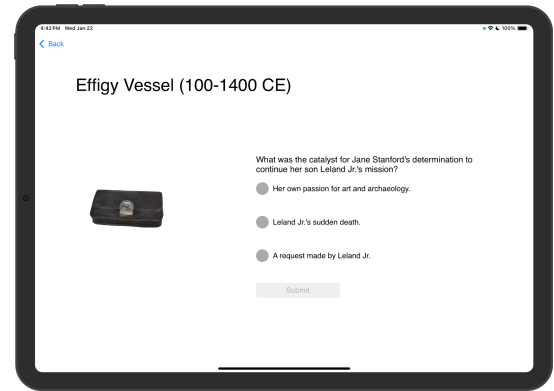
We built a high-fidelity prototype to explore the feasibility of various features, including AR artifact display, quiz questions, and a chatbot to provide feedback and answer questions that students learners might have. Based on students' age, teachers can select from different museum experiences with artifacts that can be easily linked to items commonly found in a classroom. Teachers can customize artifact descriptions and regenerate quiz questions with the help of generative AI. When students open the app, the artifacts appear right in front them. Each artifact is accompanied by a description, personalized chatbot, and trivia-type questions, where they can make observations and learn about the artifact in-depth (See Figure 1).

3.1 Artifact Curation

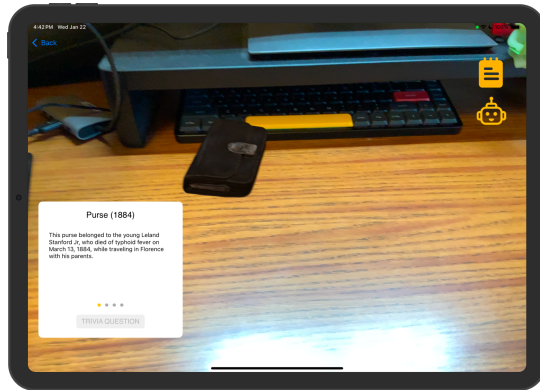
We partnered with an art museum on our university campus who helped us curate a set of artifacts from a detailed exhibit about a local historical figure. These artifacts consisted of ten unique trinkets he collected over the course of his lifetime, which were later narrowed down to six for the prototype. We scanned the artifacts using Polycam, a LiDAR and 3D scanner app, and later turned those into files that can be displayed in AR. For artifacts with unique shapes, it was challenging to scan their complete form



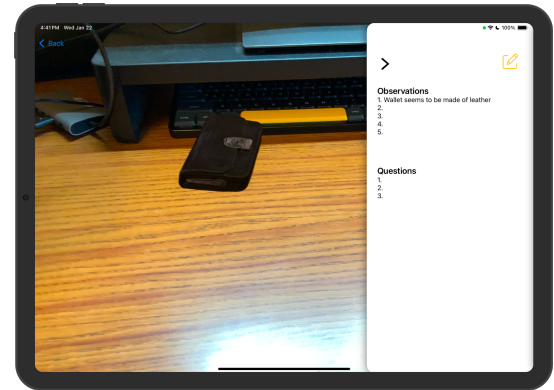
(a) Free response question for an artifact



(b) Quiz question for an artifact



(c) AR display of artifact with descriptions. Chatbot is available on the right.



(d) Students can access the note panel to view their observations and questions.

Figure 1: Overview of Museum in the Classroom.

due to their unique angles. To fill in the missing spaces, we used Photoshop to match the original shape.

3.2 App and Core Features

We developed our app in XCode using Swift, a native iOS language, and we deployed it on iPad via TestFlight. The app follows a Model-View-Controller (MVC) architecture with two primary layers: client-side and server-side components. The client-side components include the ARKit and RealityKit for AR functionality, Core Data for persistence storage of artifact information and student interactions, and Swift UI controllers that manage the teacher and student experiences. The server-side components consist of the OpenAI API integration that powers the chatbot, artifact description generation, and quiz question creation (See our system diagram in Figure 2). Our target device was a 11-inch iPad Pro, which we chose for its large screen size and its built-in LiDAR sensor for depth sensing.

In this subsection, we dive deeper into the four main features in the app: 1) artifact selection and AR display, 2) its chatbot, 3) the quiz questions, and 4) the note panel.

3.2.1 Artifact selection and AR display. This feature allows students to see artifacts in AR form by facing the camera at a flat surface.

Students are given a collection of artifacts to select from for each session. AI-generated descriptions of the artifacts were hard-coded into the app to ensure students all saw the same information. We implemented the AR view using ARKit, an iOS toolkit to build AR applications.

3.2.2 Chatbot. We designed the app such that students can interact with the chatbot to answer free-form questions about the artifacts (See Figure 1c). The chatbot was prompted to answer the questions as if it were a teacher (e.g. *“Given your question or observation about a museum artifact, the system will respond like an enthusiastic teacher with an accurate and succinct answer.”*). Our intention was to increase engagement through real-time feedback.

3.2.3 Quiz questions. There are two variations of quiz questions: multiple choice and free-response questions. We used generative AI to create the lists of multiple choice questions about each artifact. The purpose of these questions was to ensure students gain a thorough understanding of the artifacts through the interactions. For multiple choice questions, when students selected the wrong answer, the app would not provide the correct answer immediately but rather provide students additional chance to answer. In free-response questions, after students answered the questions, our interface would provide feedback based on the students’ responses.

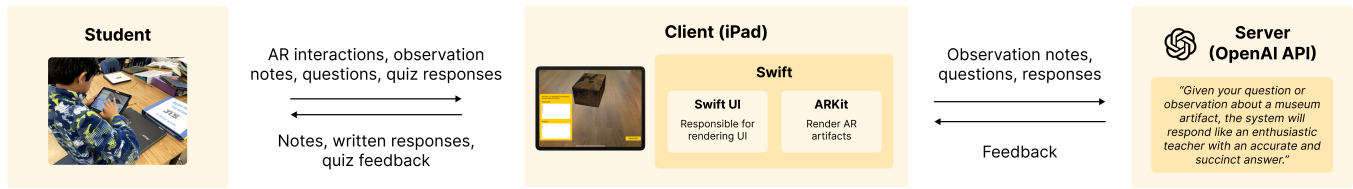


Figure 2: System diagram of Museum in the Classroom, separated by client-side and server-side components.

3.2.4 Note panel. We embedded a note section in the app for when students are interacting with AR artifacts (see Figure 1d). The note panel serves as a place for students to record observations of the artifacts, similar to traditional note-taking on paper. In contrast to paper note-taking, though, our interface would provide feedback for the student’s observation notes, as if it were a teacher. The notes recorded here were also available to students as references when for when they had to answer the free-response questions.

3.3 AI Implementation

3.3.1 OpenAI integration. The app integrates with OpenAI’s API through the OpenAIAPICaller class, which provides two primary methods: `getTeacherbotReponse` and `getGPT4Response`. `getTeacherbotReponse` calls a fine-tuned GPT-3.5 model specifically trained for educational responses, and `getGPT4Response` calls the standard GPT-4 model for more complex content generation.

3.3.2 Fine-tuning. To ensure age-appropriate responses and to reduce the risks of hallucination, we used a custom fine-tuned GPT-3.5 trained specifically for educational purposes about museum artifacts. Our fine-tuning process involved 1) creating 69 examples of teacher-like responses about museum artifact using GPT-4, 2) developing a specialized system prompt that guides the model’s behavior, and 3) training the model using OpenAI’s fine-tuning API. Domain-specific fine-tuning can significantly reduce hallucinations in specialized contexts [18]. Additionally, we used structured prompts that ground the model in the specific information about each artifact in order to provide context that can help reduce hallucination rates [22]. Our system also allows the teacher to review and edit AI-generated artifact descriptions and quiz questions before they are shown to students.

4 Evaluation

We conducted a formative study of the MITC app ($N = 20$) in which participants completed reading and learning activities using the system. This section describes our participants, procedure, and measures.

4.1 Participants

A class of seventh graders ($N = 20$: 12 female, 8 male) was recruited through word of mouth and cold outreach to local history classrooms, and the selected middle school leadership signed a letter of agreement to approve the study. We sought the informed consent of all participants’ parents or guardians and received assent from the participants at the beginning of the study. Nearly all participants (n

$= 19$) completed both sessions.¹ Participants were compensated for their time with a \$20 Amazon gift card and they were supervised by the researchers and teacher throughout the study.

4.2 Procedure

We went to the participants’ local middle school for two approximately 90-minute sessions spaced two days apart. In each session, all students were assigned a number and independently participated at the same time in the same classroom. During the first session, participants were first asked to fill out a 10-minute, 4-question pre-survey on their chromebooks. Then, participants were given an iPad or a paper packet and learned about six artifacts (Amphora, Gold Spike Case, Incense Box, Purse, Watch, and Jewel Casket) for 25 minutes either by examining them in the MITC app (treatment) - where they were encouraged to take notes via the notepad within the app - or by reading about them on a printed handout (control) - where they were encouraged to take notes on these artifacts via a Google Form. The participants with iPads turned on the screen recording feature so qualitative assessments could be made about their reactions and maneuvers within the app. In the second session, the research team returned to the classroom to release a delayed 45-minute, 4 multiple-choice/3 essay question follow-up assessment (with students’ notes given back to them) two days after the first session to assess knowledge retention. Afterward, we administered a 15 minute 9-question post-survey session to gauge engagement.

4.2.1 Condition Assignment. Participants were assigned to one of two experimental groups: the “Odd” group ($n = 10$), who engaged with the Amphora, Gold Spike Case, and Incense Box using the app, and the Purse, Watch, and Jewel Casket through static written material. The other experimental group was deemed the “Even” group ($n = 10$) and engaged with the same artifacts but vice versa. They engaged with the Amphora, Gold Spike Case, and Incense Box using the static written material and the Purse, Watch, and Jewel Casket through the app. This method allowed us to directly compare the effects within individuals. Assignments to these groups were randomized; participants counted themselves off and were assigned an odd or even number, which determined their conditions.

4.3 Measures

To gather data, we employed a pre-study survey (pre-survey form), a knowledge assessment, and a post-study survey (post-survey form).

¹One participant opted out of the post-survey during Session 2.

4.3.1 Pre- and post-survey forms. The pre-survey form collected demographic information and asked participants about their experiences with AR and technology in the classroom. We did not end up using these data in our analysis, as the intention of the pre-survey form was to have better context about the participants. Thus, we omit further mention of them. The post-survey form asked participants to share their feedback about the MITC app and evaluate their experience using MITC and the written materials during the study.

4.3.2 Knowledge assessment. We measured participants' knowledge of the artifacts and their histories through a set of assessment questions. The quiz included questions such as "What role did Luigi di Cesnola's connection to the amphora play in his career?" Participants were informed that they would not be judged for their responses. There were seven questions in total: four multiple choice questions, two short answer questions, and one essay question. The multiple choice and short answer questions each cover one of the six artifacts, while the essay question asked participants to synthesize their understanding of all the artifacts to characterize the historical owner of the artifacts. To grade the assessments, the team created a rubric for each quiz question. Points were awarded for key ideas in free-response questions and for demonstrating basic comprehension. The researchers independently scored the pre- and post-survey forms, were blinded to condition, and then came to a consensus on the final scores. Because the historical topic covered by our artifacts were new to the participants, we assume that they did not have prior knowledge. Hence, to measure learning gains, we only assess their knowledge after the experience.

To determine whether the use of the MITC app leads to higher learning gains than the written materials (i.e., the control), we divided the six artifact-specific questions into two groups: Group 1 (Amphora, Gold Spike Case, Incense Box) and Group 2 (Purse, Watch, Jewel Casket). Since those in the Odd condition learned about the Group 1 artifacts in the MITC app and those in the Even condition learned about the same artifacts with written materials, we can compare their scores on the Group 1 questions to evaluate whether there is a difference. The same logic applies vice versa for Group 2 questions.

5 Preliminary Findings

We performed a qualitative analysis of the post-survey data and organize our findings across three categories: engagement, museum visit comparisons, and AI interactions. We also ran a quantitative analysis of the knowledge assessment. There was one participant who did not complete the pre- and post-survey; their data was excluded from pre/post analyses but included in qualitative observations.

5.1 Engagement

In the post-survey, we found that most participants ($n = 17$, 89.5%) were interested in continuing to use the MITC app. Participants also generally expressed a preference for the app over the textbook, with 12 (63.2%) indicating that they preferred to history topics with the app and seven (36.8%) indicating a preference for the textbook.

We also categorized participants' responses to the question, "What was your favorite part about this experience?" After discarding generic answers, such as "the app", "being able to try out the new app and seeing how it works" or "the augmented reality", we found that participants tended to mention either the AR features or AI features of the system in their response. Eight participants (42.1%) highlighted the AR and 3D objects as their favorite. P11 noted their immersion: "I really liked how detailed the pot model was. It was like it was really there." Three participants (15.6%) mentioned the AI in their responses, referring to the chatbot in the app. For instance, P17 loved she "can be answered by an educated AI".

5.2 Comparisons to Real Museum Visits

One question on our post-survey asked participants to compare their experience with MITC to visiting a museum. Overall, the majority of participants ($n = 10$, 52.6%) stated a preference for real museums over the app. Among the remaining participants, some ($n = 7$, 36.8%) expressed a preference for the app over museums, and a few ($n = 2$, 10.5%) found them to be similar or equal.

For those who enjoyed the app more, one key distinguishing feature was being able to get up close to the artifacts. P4 remarked, "you could get a close look at the artifact and look at all the designs and you won't be able to experience that a museum because [they] have a glass border." Most participants noted that the app made the exhibit experience more interactive and consolidated all the information they needed in one place, which helped them learn better: "The app answering your questions while the museum usually just puts a sign to say the information" (P14). Some participants also appreciated that the app let them engage with the artifacts in a more comfortable environment than a busy museum, "without the pushing and shoving of the crowds" (P1).

Those who preferred visiting museums gave a variety of reasons, such as museums having a wider variety of artifacts and exhibits. Notably, two participants commented on the social and kinesthetic advantages of visiting a museum compared to using the MITC app: "[At] the museum you can be with your friends" (P7) and "it is nice to walk around with people, not just sit one [sic] your seat" (P16).

5.3 Feedback on AI Interactions

In the post-survey, we found that participants demonstrated varying perspectives on the AI-powered interactions in the app. When asked about their experiences with quiz questions, four participants stated explicitly that they appreciated AI for providing immediate feedback. While we observed an overall positive sentiment toward the chatbot, few participants had specific complaints. One was about how the AI chatbot spoke, especially around the use of indirect and hedging language. P3 stated that they learned how "vague a AI robot thing can be," bolstering P14's comments that they wish the "bot could be more direct." Another complaint was that "if you don't ask the right questions, or don't make certain observations, then you might not get the information you need" (P1).

5.4 Qualitative Observations

During the study, we noticed that some participants' body language and expressions conveyed excitement to see artifacts appear in AR.

For example, when the system situated artifacts under tables instead of on top of them (a bug that we did not intend), participants often knelt down to find the artifacts, keen to discover where the artifact was, as if they were participating in a treasure hunt. Some students were quite vocal, announcing their excitement or frustration with the app out loud to their peers ("Why are the chatbot's answers so generic?") while interacting with it. We also noticed that students would often help each other with app navigation and compare artifact location and findings with each other in order to better navigate the app. The extent of groupwork was limited to the students in close proximity, as desks were arranged in groups of four/five. Throughout the study, students were generally curious and animated, with a steady stream of reactions and conversation while the study was going on.

5.5 Learning Comparison

We did not observe any significant increase in learning outcomes due to the MITC app compared to the written materials (i.e., the control). See Section 4.3.1 for an explanation of the measures used in this section.

To evaluate the effect of the app on learning outcomes, we use two independent variables: the sum of a participant's scores on Group 1 questions (range: [0, 5]) and on Group 2 questions (range: [0, 5]). We then check for differences between the two conditions (Even and Odd) using an unpaired t-test.

For the Group 1 questions, where those in the Odd condition studied the corresponding artifacts in the app and those in the Even condition used written materials, we found no significant difference between Odd ($M = 2.2, SD = 1.55$) and Even ($M = 2.9, SD = 1.66$). For the Group 2 questions, where those in the Odd condition studied the corresponding artifacts with written materials and those in the Even condition used the app, we found no significant difference between Odd ($M = 2.9, SD = 0.88$) and Even ($M = 3.5, SD = 1.18$).

6 Discussion

Through the process of designing, implementing, and evaluating Museum in the Classroom, we aimed to better understand how to bring elements of the museum learning experience into the classroom through the use of augmented reality and generative AI.

In our formative study, we did not observe differences in learning between using MITC and using written materials—likely due to our small sample size and the limited scope of our intervention—but we did find that MITC created an environment where students could engage more deeply with museum artifacts. Alongside each artifact, MITC offered multiple ways for students to deepen their knowledge: quiz questions to test understanding, a space for notes (and immediate feedback on those notes), and a chatbot to discuss with. However, we did find room for improvement regarding clarity and usability, especially for the features where students interacted more directly with the AI, as seen in cases where participants were not sure how to ask the right questions to the chatbot. To address this "capability gap" that is commonly encountered when end users interact with large language models [30], we could add more scaffolding, such as generating example questions or giving students more practice with chatbots.

MITC also fostered a playful and comfortable environment for museum learning. The learning experience was facilitated in a familiar space—the classroom—where students would not have to deal with crowds or other distractions. (Or worry about breaking any priceless artifacts!) With virtual AR artifacts, students could enjoy getting up close to the artifacts and observing them from all angles. Their placement in the real world gave students a chance to hunt for them on the floor and under tables, which sparked a sense of discovery. We see much potential to extend this playfulness by further leveraging the affordances of augmented reality. For example, we could let students to move artifacts around to curate and organize their own personal exhibits in the classroom, giving them the opportunity to step into the role of a museum curator.

Crucially, we would like to emphasize that MITC is not intended to replace museum visits. Rather, it is meant to offer a scalable approach to be used when visiting a museum is not feasible. In fact, one avenue for future work is to consider how to incorporate physical museum visits and in-classroom virtual museum artifacts *in tandem*. Because reflection and discussion play an important role in helping students consolidate knowledge after a field trip [5], teachers could use AR in the classroom to allow students to look at the same artifacts from a different perspective.

6.1 Limitations and Future Work

The MITC system presented in this paper is still an early prototype, and we plan to continue to iterate on its design and implementation. For example, the AR artifacts in the system were chosen because they had similar dimensions or purposes to common items that could be found in the classroom. In our next prototype, we plan to have these virtual artifacts be directly overlaid on top of a corresponding physical object using AR anchors and object tracking. This would allow students to more directly interact with the AR artifacts, which addresses a complaint that our participants had. Furthermore, the teacher could spread objects out around the classroom to encourage students to move around and discover the artifacts, which would bring in an additional kinesthetic aspect akin to walking around a museum. Students could even physically manipulate the artifacts via the physical objects, which would not be possible with the original artifacts at the museum.

On the AI side, a major risk of integrating LLMs into educational experiences is the risk of these models hallucinating false information [2, 23]. While we took steps to mitigate these risks, such as fine-tuning, structured prompting with context, and human review, they do not eliminate the risk of hallucination, especially when students can freely interact with the AI, such as with a chatbot. We would like to integrate more measures to reduce hallucination, like using retrieval-augmented generation [15] to ground the model's responses in authoritative source material. We also want to introduce more safety measures for teachers to be able to monitor the LLM's responses for appropriateness and intervene if necessary.

Our vision for MITC also includes much greater involvement on behalf of the teacher. During the needfinding stage of our design, we spoke with three teachers to better understand what they are looking for when integrating new technologies into the classroom. A recurring theme that emerged was the lack of customization in the current curriculum. The teachers we talked to expressed concerns

that if certain topics were missed due to time constraints, they often had to either omit them entirely or inefficiently restructure the lesson plans so that they can stay on schedule.

To address this, the app has an existing, built-in personalized onboarding process, in which users respond to a series of targeted questions that help determine the specific curriculum needs, such as the students' age group and the historical eras being covered in class. The app uses this information to curate relevant artifacts from our ideally expansive artifact library with appropriate vocabulary for artifact descriptions. This offers a more tailored experience that aligns with each teacher's unique educational goals for their students. Furthermore, we hypothesize that this would help students engage with the artifacts more akin to how they would during a real museum visit, while reducing the "novelty space" that accompanies those visits, which has been shown to inhibit learning [27].

For the present study, we limited our scope to assess the students' experience with the app. Evaluating the long-term impacts on knowledge retention, a crucial factor for testing the effectiveness of educational interventions, was outside the scope of this study. Furthermore, we recognize that the difference in engagement that we observed in the study may be due to novelty effect or the limitations of self-report engagement measures. Additionally, in future studies, we plan to use a validated scale like the User Engagement Scale (UES) [28] to measure engagement.

Finally, we plan to work more closely with teachers and test out the onboarding process to get a more complete picture of how MITC fits into their curriculum as part of a formal evaluation for the next iteration of MITC.

7 Conclusion

Museum in the Classroom explores the potential of enhancing student engagement with historical artifacts through AR and generative AI. Our findings indicate that MITC fosters increased student enthusiasm compared to traditional learning methods, as seen through their high levels of engagement with AR artifacts and an AI-powered chatbot. However, our analysis did not reveal significant differences in learning outcomes between students using MITC and those relying on written materials, suggesting that the impact on knowledge retention requires further investigation. In future research, there is a need to include more explicit guidance on AI interactions and instructional scaffolding. Additionally, optimizing AI-generated feedback and question generation could better support student comprehension and retention.

While this app is not intended to replace museum visits, it does present a scalable alternative for schools where a field trip is not possible. By honing on the technological capabilities and more closely aligning them with teachers' learning objectives, MITC has the potential to complement traditional learning methods and provide an engaging, interactive approach to classroom education.

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